Web Technologies

Bootstrap

Summary of Today's Lecture

- Bootstrap Introduction
- Bootstrap Grid System
- Bootstrap Grid Classes
- Bootstrap Grid Example (Photo Gallery)
- Bootstrap Column Layouts (Examples)
- Bootstrap Components (Examples)
  - Navbar
  - Jumbotron
  - Breadcrumb
  - Tables
  - Buttons
  - Forms
- Responsive Web Design Example (Restaurant Web Portal)

What is Bootstrap

- **Bootstrap** is a free and open-source CSS framework directed at responsive, mobile-first front-end web development.
- It contains CSS- and (optionally) JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.
- It is a powerful front-end framework that helps create websites faster and easier.
- It was released as an open source product in August 2011 on GitHub.

What is Grid

- In graphic design, a grid is a structure (usually two-dimensional) made up of a series of intersecting straight (vertical, horizontal) lines used to structure content.
- Bootstrap includes a responsive, mobile first grid system that appropriately scales up to 12 columns as the device or viewport size increases.
- Bootstrap 4 is mobile first in the sense that the code for Bootstrap now starts by targeting smaller screens like mobile devices, tablets, and then “expands” components and grids for larger screens such as laptops, desktops.

How do grid systems work?



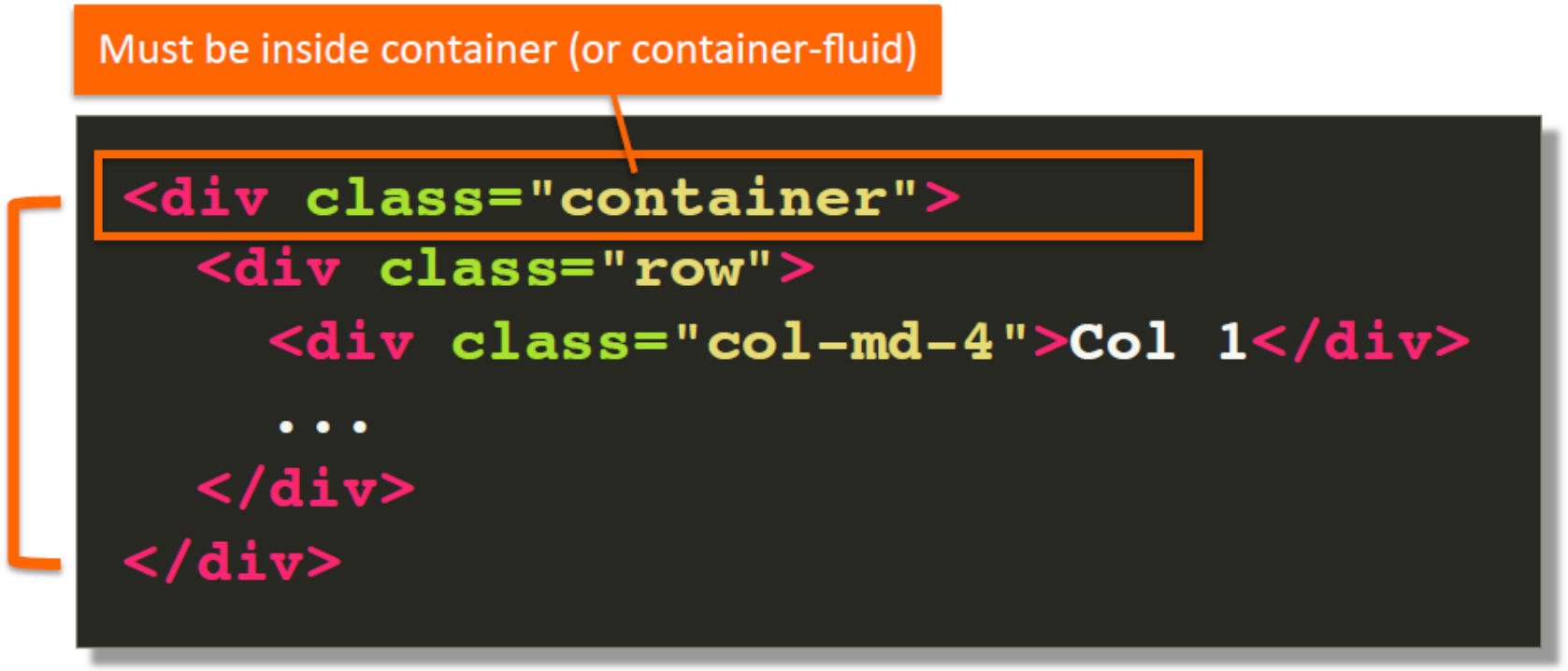
Adding Bootstrap Framework

- Two ways of adding boostrap framework to the HTML file,
- Bootstrap CDN (Content Delivery Network)
  - `<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/4.4.1/css/bootstrap.min.css">`
- Add as a separate file.
  - `<link rel="stylesheet" href="bootstrap.min.css">`

Working of Bootstrap grid System

- Rows must be placed within a **`.container`** class for proper alignment and padding.
- Use rows to create horizontal groups of columns.
- Content should be placed within columns, and only columns may be immediate children of rows.
- Predefined grid classes like **`.row`** and **`.col-sm-4`** are available for quickly making grid layouts.
- Grid columns are created by specifying the number of twelve available columns you wish to span. For example, three equal columns would use three **`.col-sm-4`**.

Container



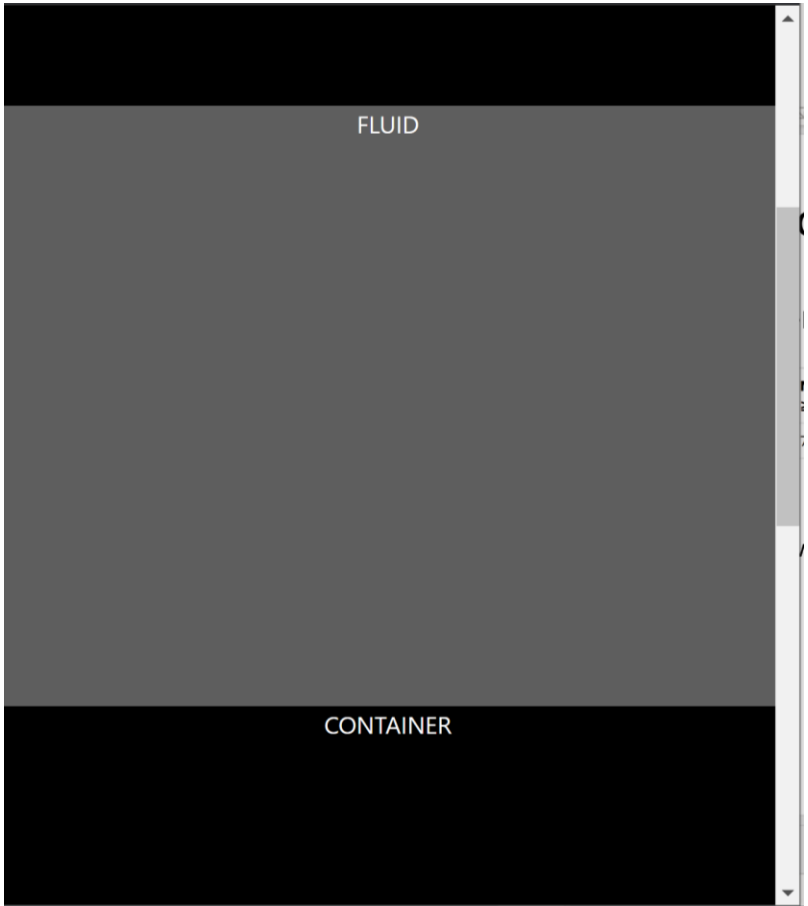
`<div class="container"> … </div>`
element is added to ensure proper centering and width for layout.

.container vs .container-fluid

- The .container class provides a responsive fixed width container.

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	Extra large ≥1200px
max-width	100%	540px	720px	960px	1140px

- The .container-fluid class provides a full width container, spanning the entire width of the viewport (width is always 100%).



.container vs .container-fluid



Activate Windows
Go to Settings to activate Windows.

Row

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Col 1</div>
    ...
  </div>
</div>
```

Once container is added, next you need think in terms of rows.
Add `<div class="row">... </div>` inside container.

Columns

```
<div class="container">
  <div class="row">
    <div class="col-md-4">Col 1</div>
    ...
  </div>
</div>
```

Every `row` in the grid is made up of `12` units and you can define the desired size of your columns using those units.
In our example we have defined one column each `4` units wide.

Grid Classes

- The Bootstrap `4` grid system has five classes:
  - `.col-` (extra small devices - screen width less than 576px)
  - `.col-sm-` (small devices - screen width equal to or greater than 576px)
  - `.col-md-` (medium devices - screen width equal to or greater than 768px)
  - `.col-lg-` (large devices - screen width equal to or greater than 992px)
  - `.col-xl-` (xlarge devices - screen width equal to or greater than 1200px)
- The classes above can be combined to create more dynamic and flexible layouts.

Responsive breakpoints

- Bootstrap is developed to be mobile first, we use a handful of media queries to create sensible breakpoints for our layouts and interfaces.
- These breakpoints are mostly based on minimum viewport widths and allow us to scale up elements as the viewport changes.

Copy

```
// Extra small devices (portrait phones, less than 576px)
// No media query for `xs` since this is the default in Bootstrap

// Small devices (landscape phones, 576px and up)
@media (min-width: 576px) { ... }

// Medium devices (tablets, 768px and up)
@media (min-width: 768px) { ... }

// Large devices (desktops, 992px and up)
@media (min-width: 992px) { ... }

// Extra large devices (large desktops, 1200px and up)
@media (min-width: 1200px) { ... }
```

Basic Structure of Bootstrap Grid

-
  -
    -
  -
  -
    -
    -
    -
  -
  -
    ...
  -
-

Extra Small Grid Example

	Extra small	Small	Medium	Large	Extra Large
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`
Screen width	<576px	>=576px	>=768px	>=992px	>=1200px

```
<div class="container">
  <div class="row">
    <div class="col-3 bg-success">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-9 bg-warning">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



Small Grid Example

	Extra small	Small	Medium	Large	Extra Large
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`
Screen width	<576px	>=576px	>=768px	>=992px	>=1200px

```
<div class="container">
  <div class="row">
    <div class="col-sm-3 bg-success">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-9 bg-warning">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



Medium Grid Example

	Extra small	Small	Medium	Large	Extra Large
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`
Screen width	<576px	>=576px	>=768px	>=992px	>=1200px

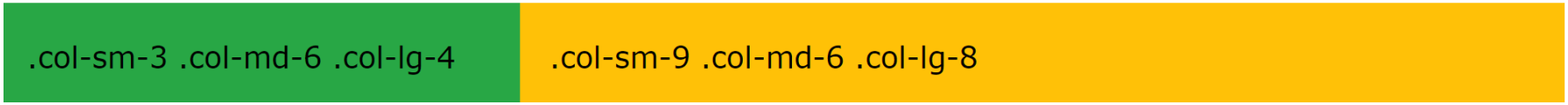
```
<div class="container">
  <div class="row">
    <div class="col-sm-3 col-md-6">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-9 col-md-6">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



Large Grid Example

	Extra small	Small	Medium	Large	Extra Large
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`
Screen width	<576px	>=576px	>=768px	>=992px	>=1200px

```
<div class="container">
  <div class="row">
    <div class="col-sm-3 col-md-6 col-lg-4">
      <p>Lorem ipsum...</p>
    </div>
    <div class="col-sm-9 col-md-6 col-lg-8">
      <p>Sed ut perspiciatis...</p>
    </div>
  </div>
</div>
```



Extra Large Grid Example

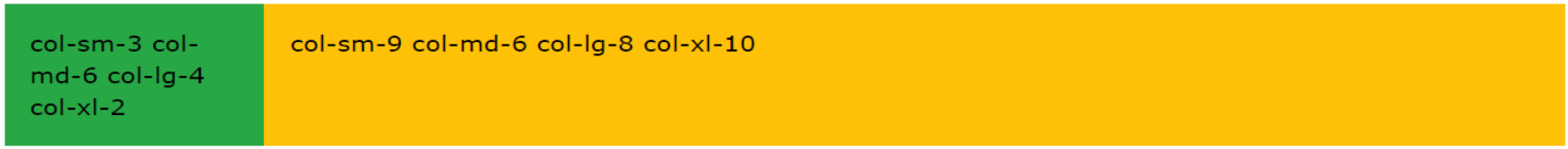
	Extra small	Small	Medium	Large	Extra Large
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`
Screen width	<576px	>=576px	>=768px	>=992px	>=1200px

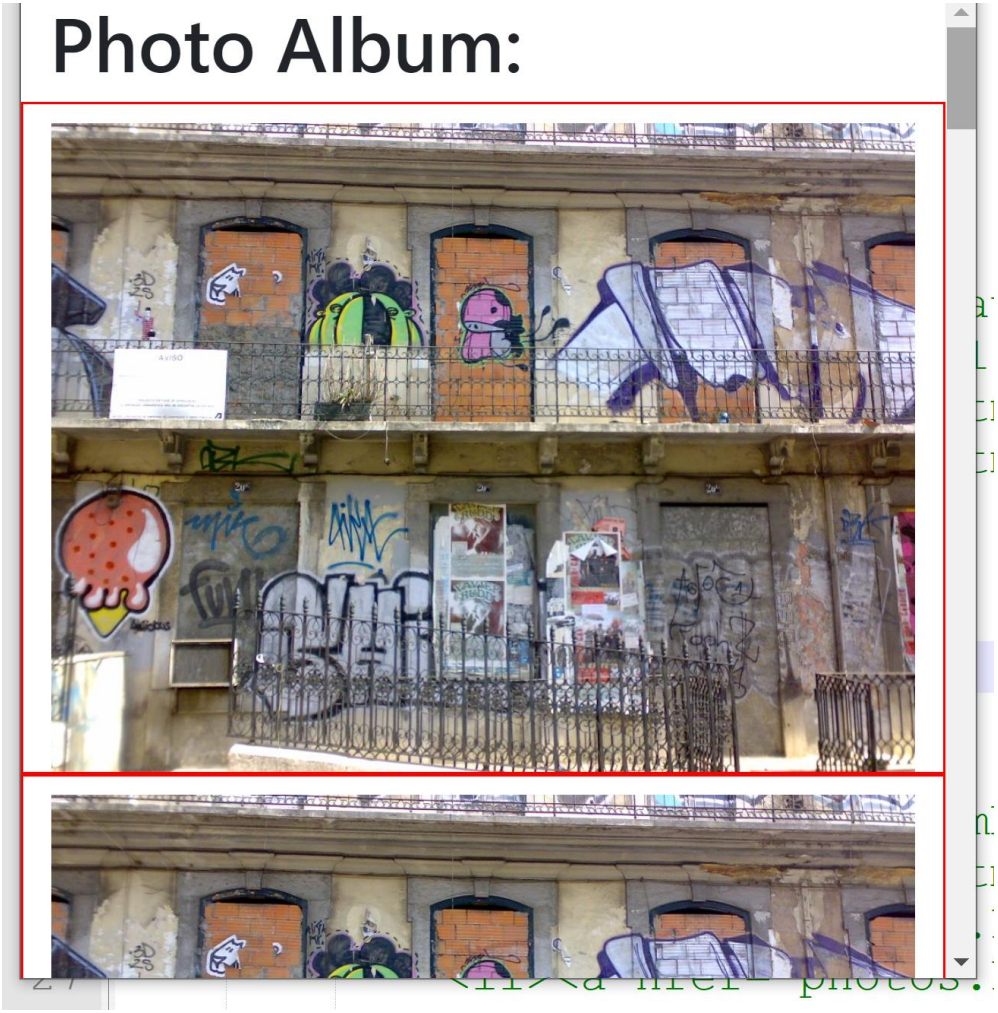
- ```

        Lorem ipsum...

        Sed ut perspiciatis...

  ```


  ```



-
-



- `<div class="col-12 col-sm-6 col-md-4 col-lg-3 col-xl-2">`
- `<a href="photo1.html"><img class="photo_list" src="./images/img_1.jpg" alt="graffiti cover image"/></a></div>`

Photo Album:



- `<div class="col-12 col-sm-6 col-md-4 col-lg-3 col-xl-2">`
- `<a href="photo1.html"><img class="photo_list" src="./images/img_1.jpg" alt="graffiti cover image"/></a></div>`

Photo Album:



- `<div class="col-12 col-sm-6 col-md-4 col-lg-3 col-xl-2">`
- `<a href="photo1.html"><img class="photo_list" src="./images/img_1.jpg" alt="graffiti cover image"/></a></div>`

Photo Album:

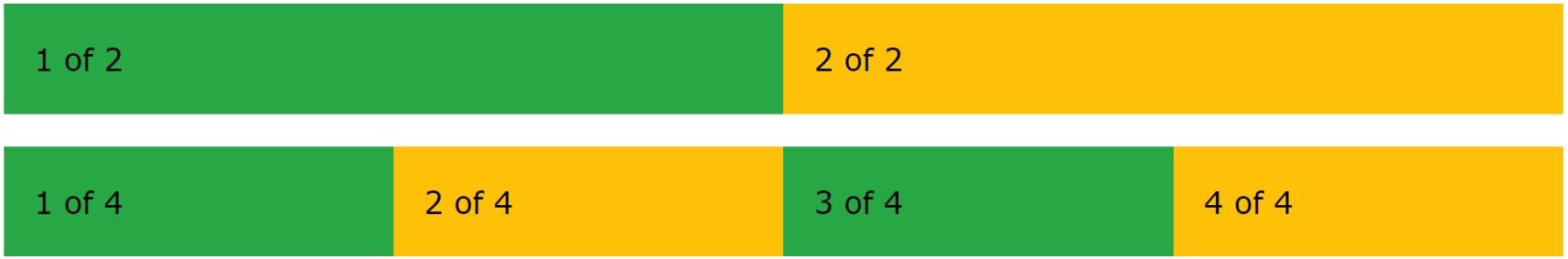


- `<div class="col-12 col-sm-6 col-md-4 col-lg-3 col-xl-2">`
- `<a href="photo1.html"><img class="photo_list" src="./images/img_1.jpg" alt="graffiti cover image"/></a></div>`

Example (Auto Column Layout)

```
<!-- Two columns: 50% width-->
<div class="row">
  <div class="col">1 of 2</div>
  <div class="col">2 of 2</div>
</div>

<!-- Four columns: 25% width-->
<div class="row">
  <div class="col">1 of 4</div>
  <div class="col">2 of 4</div>
  <div class="col">3 of 4</div>
  <div class="col">4 of 4</div>
</div>
```



Example (Three equal columns)

- Get three equal-width columns **starting at desktops and scaling to large desktops**.
- On mobile devices, tablets and below, the columns will automatically stack.
- `<div class="container">`
  - `<div class="row">`
    - `<div class="col-md-4" style="background-color:lavender;">.col-md-4</div>`
    - `<div class="col-md-4" style="background-color:lavenderblush;">.col-md-4</div>`
    - `<div class="col-md-4" style="background-color:lavender;">.col-md-4</div>`
  - `</div>`
- `</div>`

Example (Three unequal Columns)

-
  -
    - .col-md-3
    - .col-md-6
    - .col-md-3
  -
-

Example (Two nested columns)

-
  -
    - .col-md-8
      -
        - .col-md-6
        - .col-md-6
      -
    -
    - .col-md-4
  -
-

Example (Column breaks)

```
<div class="row">
  <div class="col-6 col-sm-3">.col-6 .col-sm-3</div>
  <div class="col-6 col-sm-3">.col-6 .col-sm-3</div>

  <!-- Force next columns to break to new line -->
  <div class="w-100"></div>

  <div class="col-6 col-sm-3">.col-6 .col-sm-3</div>
  <div class="col-6 col-sm-3">.col-6 .col-sm-3</div>
</div>
```

.col-6 .col-sm-3	.col-6 .col-sm-3
.col-6 .col-sm-3	.col-6 .col-sm-3

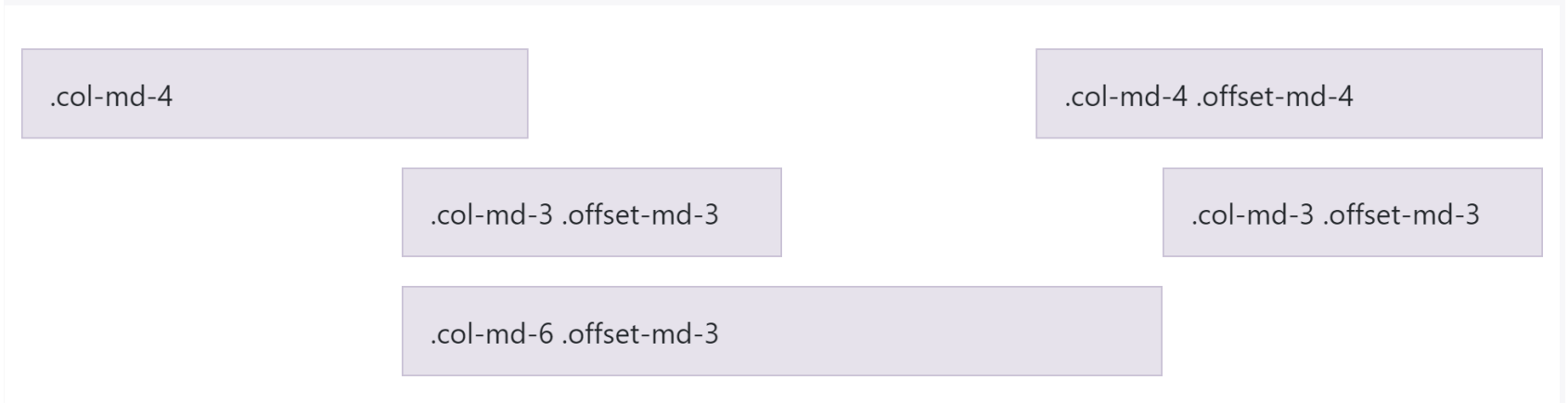
Example (Offsetting Column)

- `<div class="container">`
  - `<div class="row">`
    - `<div class="col-sm-5 offset-sm-2 col-md-6 offset-md-0">.col-sm-5 .col-sm-offset-2 .col-md-6 .col-md-offset-0</div>`
    - `<div class="col-sm-5 col-md-6">.col-sm-5 .col-md-6</div>`
  - `</div>`
- `</div>`



Example (Offsetting Column)

```
<div class="row">
  <div class="col-md-4">.col-md-4</div>
  <div class="col-md-4 offset-md-4">.col-md-4 .offset-md-4</div>
</div>
<div class="row">
  <div class="col-md-3 offset-md-3">.col-md-3 .offset-md-3</div>
  <div class="col-md-3 offset-md-3">.col-md-3 .offset-md-3</div>
</div>
<div class="row">
  <div class="col-md-6 offset-md-3">.col-md-6 .offset-md-3</div>
</div>
```



Bootstrap Jumbotron

- A jumbotron is used to showcase key content on a website.
- It indicates a big grey box for calling extra attention to some special content or information.

```
<div class="container">
  <div class="jumbotron">
    <h1>Bootstrap Tutorial</h1>
    <p>Bootstrap is the most popular HTML, CSS, and JS framework for developing responsive, mobile-first projects on the web.</p>
  </div>
  <p>This is some text.</p>
  <p>This is another text.</p>
</div>
```



This is some text.

This is another text

Bootstrap Jumbotron

```
<div class="jumbotron">
  <div class="container">
    <h1>Bootstrap Tutorial</h1>
    <p>Bootstrap is the most popular HTML, CSS, and JS framework for developing responsive, mobile-first projects on the web.</p>
  </div>
</div>

<div class="container">
  <p>This is some text.</p>
  <p>This is another text.</p>
</div>
```



Bootstrap Buttons

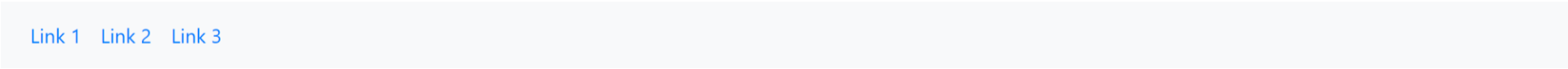
-
  - <button type="button" class= "btn btn-primary">hello click me</button>
  - <button type="button" class= "btn btn-seondary">hello click me</button>
  - <button type="button" class= "btn btn-success">hello click me</button>
  - <button type="button" class= "btn btn-danger">hello click me</button>
  - <button type="button" class= "btn btn-info">hello click me</button>
  - <button type="button" class= "btn btn-light disabled">hello click me</button>
  - <button type="button" class= "btn btn-outline-danger btn-sm">hello click me</button>
-

Buttons groups

```
<div class="container">
  <h2>Button Group</h2>
  <p>The .btn-group class creates a button group:</p>
  <div class="btn-group">
    <button type="button" class="btn btn-primary">Apple</button>
    <button type="button" class="btn btn-primary">Samsung</button>
    <button type="button" class="btn btn-primary">Sony</button>
  </div>
</div>
```

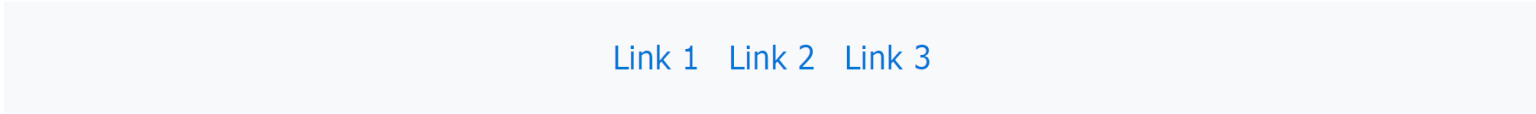
Bootstrap Navbar

```
<nav class="navbar navbar-expand-sm bg-light">
  <ul class="navbar-nav">
    <li class="nav-item">
      <a class="nav-link" href="#">Link 1</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link 2</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" href="#">Link 3</a>
    </li>
  </ul>
</nav>
```

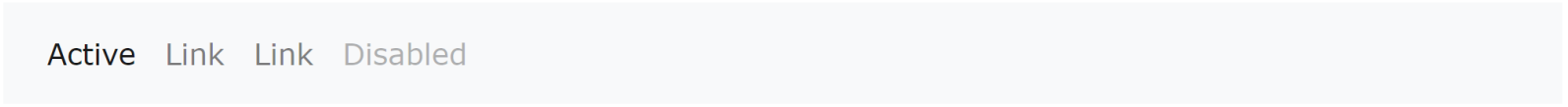


Navbar Options

- `<nav class="navbar navbar-expand-sm bg-light justify-content-center">`
  `...`
  `</nav>`



- `<nav class="navbar navbar-expand-sm bg-dark navbar-dark">...</nav>`
- Use any of the `.bg-color` classes to change the background color of the navbar (`.bg-primary`, `.bg-success`, `.bg-info`, `.bg-warning`, `.bg-danger`, `.bg-secondary`, `.bg-dark` and `.bg-light`)
- Add a white text color to all links in the navbar with the `.navbar-dark` class, or use the `.navbar-light` class to add a black text color.







Navbar Options

- ```
  <nav class="navbar navbar-expand-sm bg-dark
  navbar-dark">
     <a class="navbar-brand" href="#">
      <img src="bird.jpg" alt="Logo" style="width:40
  px;"> </a>
    ...
  </nav>
  ```



- **Top Fixed Navbar**
- ```
  <nav class="navbar navbar-expand-sm bg-dark
  navbar-dark fixed-top">
    ...
  </nav>
  ```
- **Bottom Fixed Navbar**
- ```
  <nav class="navbar navbar-expand-sm bg-dark
  navbar-dark fixed-bottom">
    ...
  </nav>
  ```
  ```

Bootstrap Breadcrumb

- Breadcrumb indicates the current page's location within a navigational hierarchy.

```
<nav>
  <ol class="breadcrumb">
    <li class="breadcrumb-item active">Home</li>
  </ol>
</nav>
```

```
<nav>
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item active">Library</li>
  </ol>
</nav>
```

```
<nav>
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Library</a></li>
    <li class="breadcrumb-item active">Data</li>
  </ol>
</nav>
```

Home

Home / Library

Home / Library / Data

Bootstrap Tables

```
<table class="table">
  <thead><tr>
      <th>Firstname</th>
      <th>Lastname</th>
      <th>Email</th></tr>
  </thead>
  <tbody>
    <tr>
      <td>John</td>
      <td>Doe</td>
      <td>john@example.com</td>
    </tr>
    <tr>
      <td>Mary</td>
      <td>Moe</td>
      <td>mary@example.com</td>
    </tr>
    <tr>
      <td>July</td>
      <td>Dooley</td>
      <td>july@example.com</td>
    </tr>
  </tbody>
</table>
```

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

Bootstrap Table (Striped)

```
<table class="table table-striped">
    <thead>
      <tr>
        <th>Firstname</th>
        <th>Lastname</th>
        <th>Email</th>
      </tr>
    </thead>
    <tbody>
      <tr>
        <td>John</td>
        <td>Doe</td>
        <td>john@example.com</td>
      </tr>
      <tr>
        <td>Mary</td>
        <td>Moe</td>
        <td>mary@example.com</td>
      </tr>
      <tr>
        <td>July</td>
        <td>Dooley</td>
        <td>july@example.com</td>
      </tr>
    </tbody>
  </table>
```

Firstname	Lastname	Email
John	Doe	john@example.com
Mary	Moe	mary@example.com
July	Dooley	july@example.com

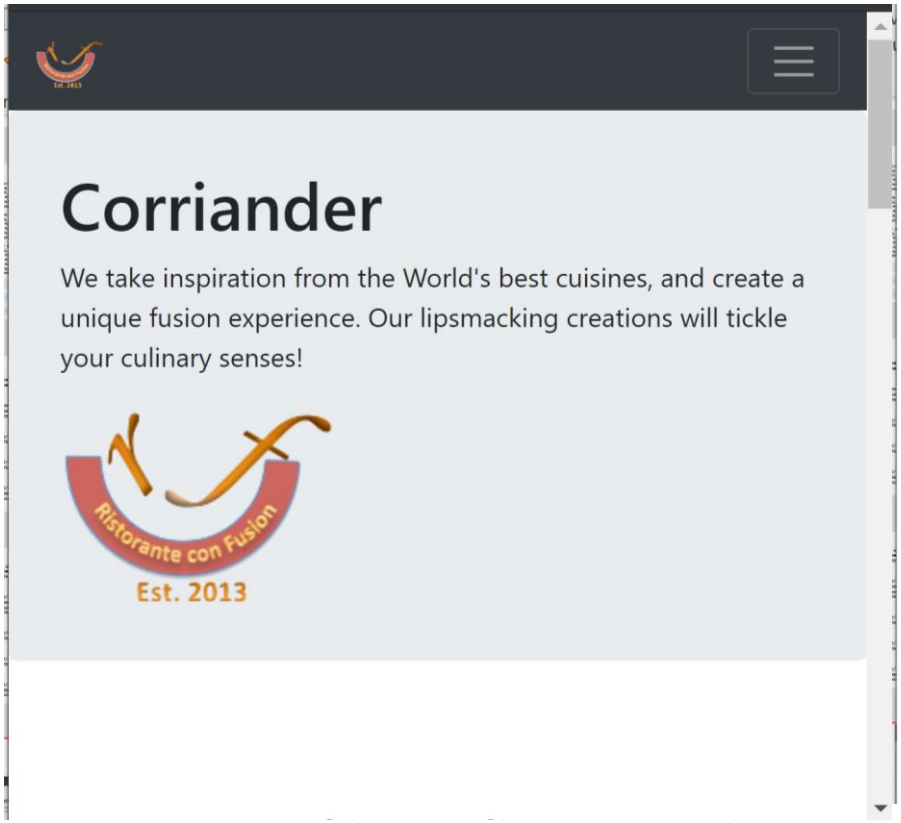
Bootstrap Forms (Stacked)

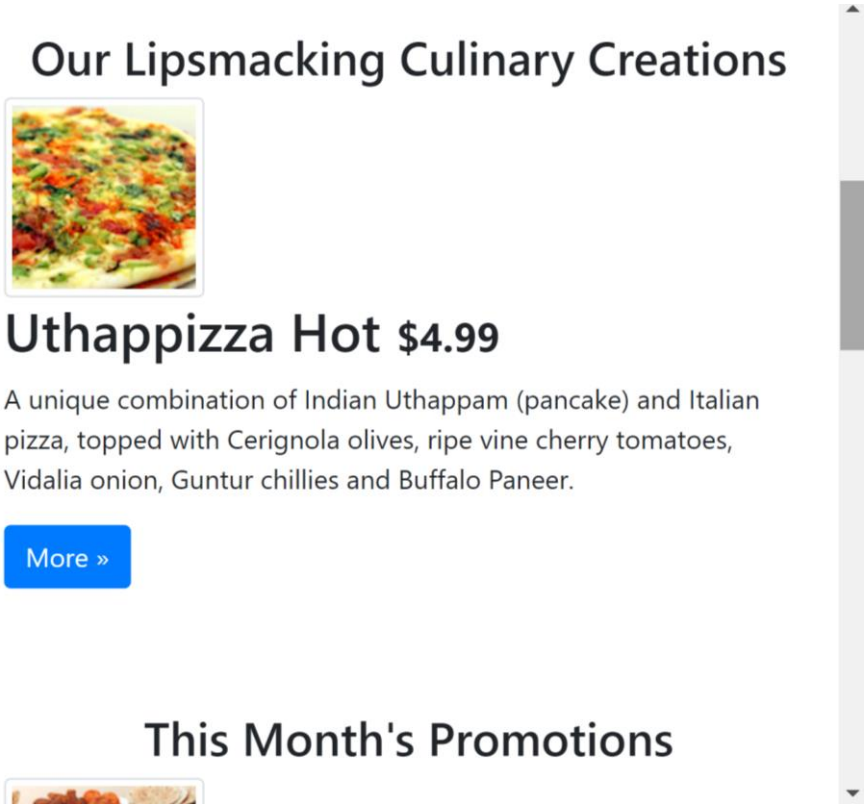
-
- <form>
  -
    - <label for="user"> studentname: </label>
    - <input type="text" name="" id="user" class="form-control">
  -
  -
    - <label for="password"> Password: </label>
    - <input type="text" name="" id="password" class="form-control">
  -
  -
    - <label for="email"> Email: </label>
    - <input type="text" name="" id="" class="form-control">
  -
  - <input type="submit" name="" class= "btn btn-success">
- </form>
-

Bootstrap Forms (Inline)

-
- <form class="form-inline">
  -
    - <label for="user"> studentname: </label>
    - <input type="text" name="" id="user" class="form-control">
  -
  -
    - <label for="password"> Password: </label>
    - <input type="text" name="" id="password" class="form-control">
  -
  -
    - <label for="email"> Email: </label>
    - <input type="text" name="" id="" class="form-control">
  -
  - <input type="submit" name="" class= "btn btn-success">
- </form>
-

Restaurant Portal (Extra-Small Screen) Home.html





Restaurant Portal (Small Screen)
Home.html

Home About Contact

Corriander

We take inspiration from the World's best cuisines, and create a unique fusion experience. Our lipsmacking creations will tickle your culinary senses!

Ristorante con Fusion
Est. 2013



Uthappizza Hot $4.99

A unique combination of Indian Uthappam (pancake) and Italian pizza, topped with Cerignola olives, ripe vine cherry tomatoes, Vidalia onion, Guntur chillies and Buffalo Paneer.

More »



Weekend Grand Buffet

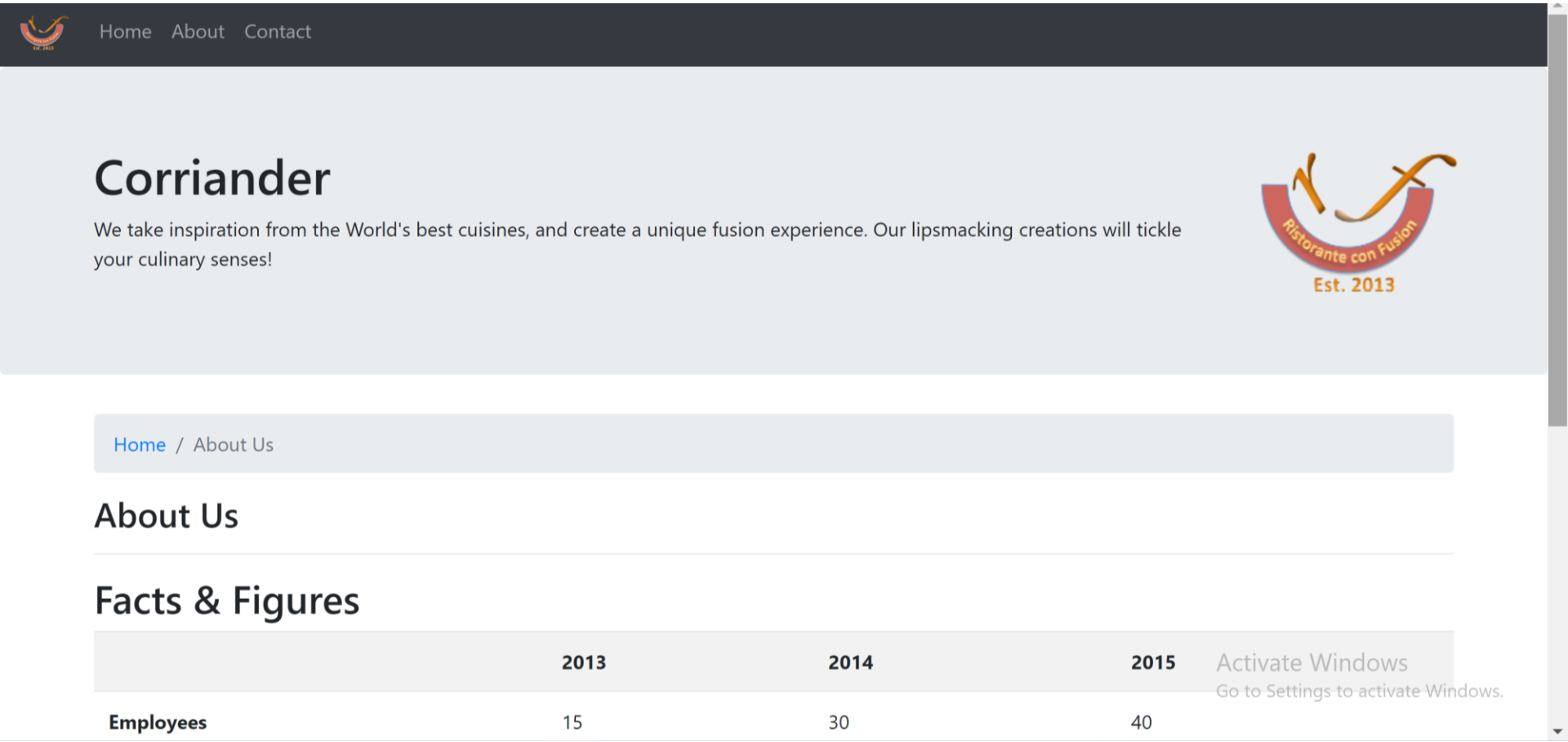
Featuring mouthwatering combinations with a choice of five different salads, six

Our Lipsmacking Culinary Creations

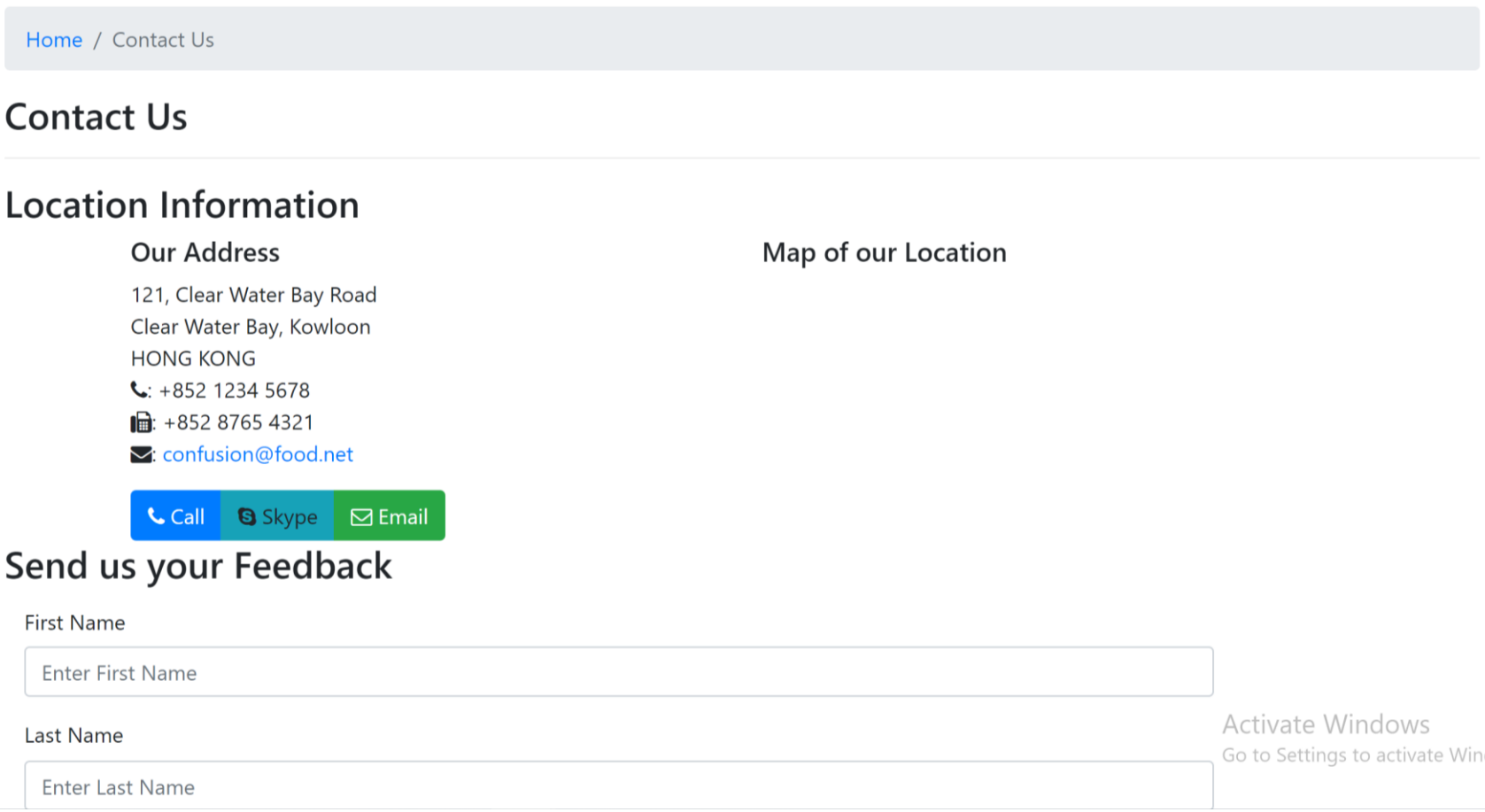
Activate Windows
Go to Settings to activate Windows.

This Month's

Restaurant Portal (Small Screen)
Aboutus.html



Restaurant Portal (Small Screen)
Contactus.html



Restaurant Portal (Small Screen)
Contactus.html

Send Feedback

Links

Home
About
Menu
Contact

Our Address

121, Clear Water Bay Road
Clear Water Bay, Kowloon
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Activate Windows
Go to Settings to activate Windows.

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Summary of Today's Lecture

- Bootstrap Introduction
- Bootstrap Grid System
- Bootstrap Grid Classes
- Bootstrap Grid Example (Photo Gallery)
- Bootstrap Column Layouts (Examples)
- Bootstrap Components (Examples)
  - Navbar
  - Jumbotron
  - Breadcrumb
  - Tables
  - Buttons
  - Forms
- Responsive Web Design Example (Cuisine Web Portal)

References

- https://getbootstrap.com/docs/4.5/getting-started/introduction/
- https://getbootstrap.com/docs/4.4/layout/grid/
- https://www.w3schools.com/bootstrap4/bootstrap_grid_system.asp
- https://en.wikipedia.org/wiki/Bootstrap_(front-end_framework)
- https://v5.getbootstrap.com/docs/5.0/getting-started/introduction/